

6. CONVOLUTION AND SAMPLING TECHNIQUES

Objective

This experiment is divided into four simulations:

1. Convolution of Continuous Time Signals
2. Convolution of Discrete Time Signals
3. Sampling Theorem Verification
4. Aliasing Effect Demonstration

Experiment 6.1 Convolution of Continuous Time Signals

Aim:

To perform the convolution of two continuous-time signals using MATLAB. The Aim is to understand the concept of convolution as a mathematical operation that combines two signals to produce a third signal, representing the amount of overlap between the two as one signal is shifted over the other.

Program:

```
% Convolution of Continuous-Time Signals
```

```
clc;
```

```
clear;
```

```
close all;
```

```
% Time axis
```

```
dt = 0.01;
```

```
t = 0:dt:5;
```

```
% Input signals  
x1 = exp(-t);      % Exponential signal  $e^{-t}$   
x2 = ones(size(t)); % Unit step signal  $u(t)$ 
```

```
% Convolution  
y = conv(x1, x2) * dt;
```

```
% Time axis for output  
t_conv = 0:dt:(length(y)-1)*dt;
```

```
% Plotting  
figure;
```

```
subplot(3,1,1);  
plot(t, x1, 'LineWidth', 2);  
grid on;  
title('Signal  $x_1(t) = e^{-t}$ ');  
xlabel('Time (s)');  
ylabel('Amplitude');
```

```
subplot(3,1,2);  
plot(t, x2, 'LineWidth', 2);  
grid on;
```

```
title('Signal x_2(t) = u(t)');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
subplot(3,1,3);
```

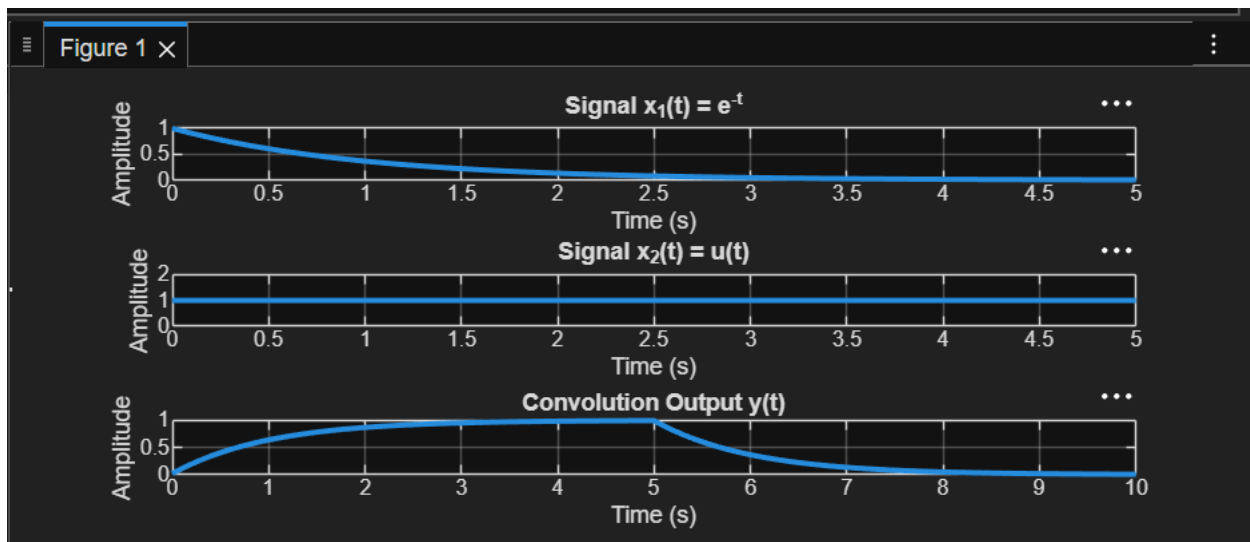
```
plot(t_conv, y, 'LineWidth', 2);
```

```
grid on;
```

```
title('Convolution Output y(t)');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude'); Output Waveform:
```



Result:

The convolution of the continuous-time signals $x_1(t) = e^{-t}u(t)$ and $x_2(t) = u(t)$ was performed successfully using MATLAB. The output signal $y(t) = 1 - e^{-t}$ was obtained and plotted, demonstrating the convolution operation for continuous-time systems.

Experiment 6.2

Convolution of Discrete Time Signals

Aim:

To compute and visualize the convolution of two discrete-time signals. Convolution is a fundamental operation in discrete-time signal processing that combines two signals to produce a third signal, which is a modified version of one of the signals.

Program:

% Convolution of Discrete-Time Signals

clc;

clear;

close all;

% Input sequences

x1 = [1 2 3 4];

x2 = [1 1 1];

% Convolution

y = conv(x1,x2);

% Sample indices

n1 = 0:length(x1)-1;

n2 = 0:length(x2)-1;

ny = 0:length(y)-1;

% Display output

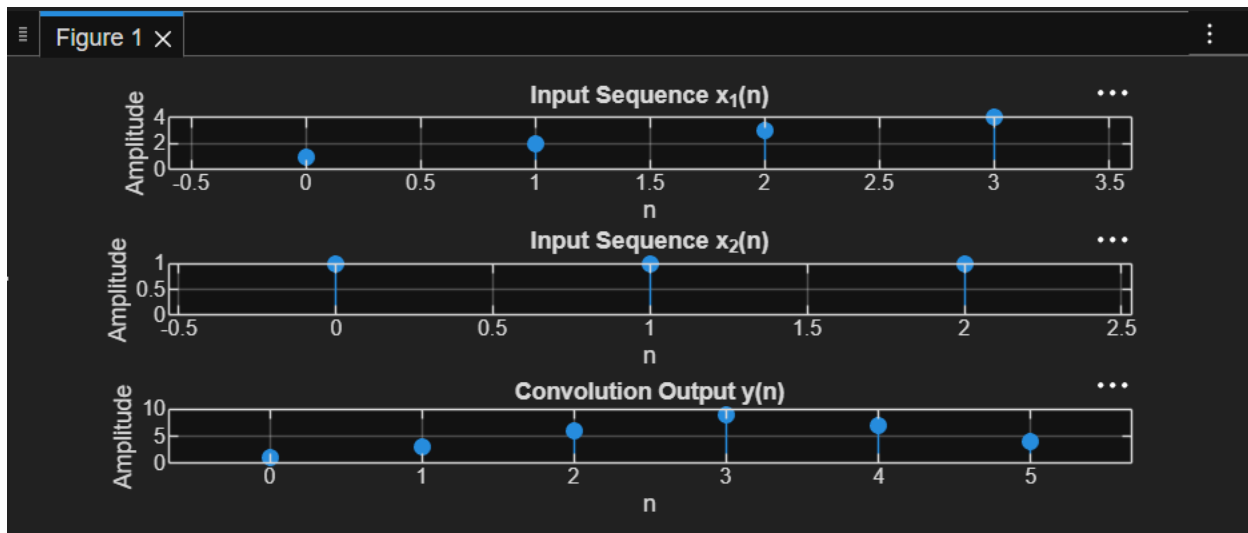
disp('Convolution Output:');

disp(y);

% Plotting

```
figure;  
subplot(3,1,1);  
stem(n1,x1,'filled');  
grid on;  
title('Input Sequence  $x_1(n)$ ');  
xlabel('n');  
ylabel('Amplitude');  
subplot(3,1,2);  
stem(n2,x2,'filled');  
  
grid on;  
title('Input Sequence  $x_2(n)$ ');  
xlabel('n');  
ylabel('Amplitude');  
subplot(3,1,3);  
stem(ny,y,'filled');  
grid on;  
title('Convolution Output  $y(n)$ ');  
xlabel('n');  
ylabel('Amplitude');
```

Output Waveform:



Result:

The convolution of the discrete-time sequences $x_1(n)=\{1,2,3,4\}$ and $x_2(n)=\{1,1,1\}$ was performed successfully using MATLAB. The output sequence obtained was $y(n)=\{1,3,6,9,7,4\}$ demonstrating the convolution operation for discrete-time signals.